Science / Space / Defence

> **Priority S** | About **8%** of Current Affairs, and rising Many questions now come as Science or Defence tags, not only "Current Affairs".

What to note every time

For space news

- Mission name, date, agency (often ISRO), rocket, and target body.

For defence news

- Weapon or system name, range, who made it (often DRDO), which force will use it, and where it was tested or shown.

High-Yield Space & Defence Foundational Spine (2024–2025 Matrix)

> **Exam Focus:** UPPCS & UKPCS Prelims regularly test ISRO landmark missions, launch vehicles, payloads, and bilateral military exercises (especially exercises conducted at high-altitude training areas in Uttarakhand like Auli and Pithoragarh).

1. ISRO Landmark Missions & Space Milestones

Mission	Launch Vehicle / Date	Orbit / Destination	Scientific Objective / Key Highlights
Chandrayaan-3	**LVM3-M4** (14 July 2023)	Lunar South Pole (Landed **23 Aug 2023**)	• Lander: **Vikram**; Rover: **Pragyan**. • Landing site named **Shiv Shakti Point**; Chandrayaan-2 crash site named **Tiranga Point**. • India became the **1st country to soft-land near lunar South Pole**, 4th country overall to land on Moon. • **23 August** declared **National Space Day**. • Payloads: ChaSTE (thermal), ILSA (seismicity), RAMBHA-LP (plasma), APXS & LIBS (chemical composition). • *Hop Experiment:* Vikram hopped 50 cm; revealed 2-layer stratigraphy in top 6.5 cm of regolith.
Aditya-L1	**PSLV-C57** (2 Sep 2023)	Halo orbit around **Sun-Earth Lagrange Point 1 (L1)** (~1.5 million km)	• India's **first dedicated solar observatory** in space. • Inserted into L1 halo orbit on **6 January 2024**. • Key payloads: **VELC** (Visible Emission Line Coronagraph - primary payload built by IIA Bengaluru), **SUIT** (Solar Ultraviolet Imaging Telescope), ASPEX, PAPA.
XPoSat (X-ray Polarimeter Satellite)	**PSLV-C58** (1 Jan 2024)	Low Earth Orbit (~650 km)	• India's first dedicated polarimetry mission to study celestial X-ray sources (black holes, neutron stars). • World's **2nd such mission** after NASA's IXPE. • Payloads: **POLIX** (RRI Bengaluru) & **XSPECT** (URSC).
Lunar Trailblazer (NASA / Caltech)	**SpaceX Falcon 9** (26 Feb 2025)	Planned lunar polar orbit (contact lost after launch)	• Smallsat mission to **detect and map water** (H_2O / OH / ice) on the Moon's surface and in permanently shadowed regions. • Instruments: **HVM3** spectrometer (JPL) and **Lunar Thermal Mapper (LTM)** (University of Oxford / UKSA). • Rideshare with Intuitive Machines **IM-2**. Mission ended **31 July 2025** after two-way contact could not be restored. • Primary objective in stems = **map lunar water**, not build a full 3D Moon model or test rovers.
Gaganyaan Programme	**Human-Rated LVM3 (HLVM3)**	Low Earth Orbit (400 km)	• 3-member crew for 3 days. • Astronaut designates announced Feb 2024: Gp Capt **Prashanth Nair**, Gp Capt **Ajit Krishnan**, Gp Capt **Angad Pratap**, Wg Cdr **Shubhanshu Shukla**. • Female humanoid robot: **Vyommitra**.

Mission	Launch Vehicle / Date	Orbit / Destination	Scientific Objective / Key Highlights
NISAR Satellite	**GSLV-F16** (30 July 2025)	Sun-Synchronous Polar Orbit (747 km circular, 98.4°)	• Landmark NASA–ISRO Earth Observation collaboration. • World's **first dual-band radar imaging satellite** using **L-band (NASA)** and **S-band (ISRO)** SAR. • Features 12-m unfurlable mesh reflector antenna on 9-m boom; **SweepSAR** technology; 12-day Earth repeat cycle; mass **2,393 kg**. • 18th GSLV flight, 12th with indigenous cryogenic upper stage.
LVM3-M6 / BlueBird Block-2	**LVM3-M6** (24 Dec 2025)	Low Earth Orbit (LEO)	• 104th launch from Sriharikota; 9th flight of LVM3 (100% success rate); 3rd dedicated commercial mission. • Launched **AST SpaceMobile (USA)** BlueBird Block-2 direct-to-cell satellite. • **Heaviest payload launched by LVM3 from Indian soil: 6,100 kg**! • Largest commercial communication satellite deployed in LEO (223 m^2 phased array antenna).

Indigenous ATP and LCH names (high-yield)

- **Kavach** is India's indigenous **Automatic Train Protection (ATP)** system for railway collision avoidance. It is **not** Rakshak / Tri-Netra / Traintel.
- **Prachand** is the IAF name of HAL's indigenous **Light Combat Helicopter (LCH)**, inducted at **Jodhpur (October 2022)**. Dhruv / Rudra / Tejas / Cheetah are **different** platforms. If a stem asks for the LCH inducted into the IAF and **Prachand is not in the options**, the key is often **None of the above**.

| **Axiom Mission 4 (Ax-4)** | **SpaceX Falcon-9**
(25 June – 15 July 2025) | International Space Station (ISS) | • Spacecraft: SpaceX Crew Dragon **'Grace' (C213)**.
• Gp Capt **Shubhanshu Shukla** (born in **Lucknow, UP**) served as **Pilot**.
• **1st Indian astronaut to visit ISS**; 2nd Indian citizen in orbit after Rakesh Sharma (1984) after 41 years.
• Commander: Peggy Whitson (USA); Crew: Tibor Kapu (Hungary), Slawosz Uznanski (Poland). Docked at Harmony Module for 18 days; splashed down in Pacific Ocean. | | **Kalam 1200 / Vikram-1** | Static Test
(8 August 2025) | SDSC-SHAR Sriharikota | • **Skyroot Aerospace** (Hyderabad) tested 1st stage motor of **Vikram-1** rocket.
• 11m long, 1.7m diameter monolithic composite motor, 30 tonnes solid propellant (longest monolithic motor built at Solid Propellant Plant, Sriharikota).
• Vikram-1 payload: 350 kg to 500 km LEO, 260 kg to SSPO. |

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2. High-Yield Bilateral & Multilateral Military Exercises (With Uttarakhand & UP Venues)

Exercise Name	Participating Nations	Arm / Service	Frequent Host Locations / High-Yield Notes
Surya Kiran	**India – Nepal**	Army	Alternates between **Pithoragarh (Uttarakhand)** & Saljhandi (Nepal). Counter-insurgency & mountain warfare. *(High-yield UKPCS hook)*
Yudh Abhyas	**India – USA**	Army	Conducted at **Auli (Chamoli, Uttarakhand)** — High-altitude cold-weather warfare. *(Crucial UKPCS hook)*
Dustlik	**India – Uzbekistan**	Army	Conducted at **Pithoragarh (Uttarakhand)** and Termez (Uzbekistan).
Mitra Shakti	**India – Sri Lanka**	Army	Alternates (e.g. Pune / Aundh).
Vajra Prahar	**India – USA**	Special Forces	Bakloh (HP) / Umroi (Meghalaya) / Idaho (USA).
Dharma Guardian	**India – Japan**	Army	Camp Imazu (Japan) / Mahajan (Rajasthan).
Garuda Shakti	**India – Indonesia**	Special Forces	Alternates between India and Indonesia.
Varuna, Garuda, Shakti	**India – France**	Navy (Varuna), Air Force (Garuda), Army (Shakti)	Bilateral multi-service exercises.

Exercise Name	Participating Nations	Arm / Service	Frequent Host Locations / High-Yield Notes
Nomadic Elephant	**India – Mongolia**	Army	Ulaanbaatar (Mongolia) / Umroi (Meghalaya).
Malabar	**QUAD (India, USA, Japan, Australia)**	Navy	Multilateral naval exercise in Indo-Pacific.

High-Yield Biotech, Computing & Space Foundation (Late 2025)

- **TOP 500 Supercomputers (66th Edition — Nov 2025, SC25 St. Louis):**
- **Global Exascale Leaders:**
  1. **El Capitan** (USA — Lawrence Livermore National Laboratory): **1.809 exaFLOP/s** (1,809 PFlop/s Rmax, 11.34 million cores; World #1).
  2. **Frontier** (USA — Oak Ridge National Lab): 1,353 PFlop/s Rmax.
  3. **Aurora** (USA — Argonne National Lab): 1,012 PFlop/s Rmax.
  4. **JUPITER Booster** (Germany — Forschungszentrum Jülich): 1,000 PFlop/s Rmax (*Europe's first exascale supercomputer*).
  5. **Eagle** (USA — Microsoft Azure): 561.2 PFlop/s Rmax.
- **India's 5 Supercomputers in Global TOP 500:**
  1. **Rank 188 — AIRAWAT-PSAI:** Located at **C-DAC, Pune** (Rmax 8.50 PFlop/s, Rpeak 13.17 PFlop/s; India's fastest supercomputer).
  2. **Rank 196 — Arka:** Located at **IITM, Pune** (Rmax 8.28 PFlop/s, Rpeak 10.61 PFlop/s).
  3. **Rank 251 — Arunika:** Located at **NCMRWF, Noida (Uttar Pradesh)** (Rmax 5.94 PFlop/s, Rpeak 7.40 PFlop/s; crucial weather forecasting system).
  4. **Rank 338 — Pratyush:** Located at **IITM, Pune** (Rmax 3.76 PFlop/s, Rpeak 4.01 PFlop/s).
  5. **Rank 469 — Arka AI/ML:** Located at **IITM, Pune** (Rmax 2.70 PFlop/s, Rpeak 3.75 PFlop/s).
- **First Indigenous CRISPR-Based Gene Therapy (19 Nov 2025):** India launched its first indigenous CRISPR Cas-based gene therapy for the clinical management of **Sickle Cell Disease**.
- **First Indigenous Antibiotic — Nafithromycin:** India's first indigenously developed and approved novel macrolide antibiotic, targeted against drug-resistant respiratory infections.
- **First Indigenous 64-Bit Microprocessor — DHRUV64 (15 Dec 2025):** India's first indigenous 64-bit dual-core microprocessor developed by **C-DAC** (Centre for Development of Advanced Computing).
- **Spiral Galaxy 'Alaknanda' (Dec 2025):** Giant spiral galaxy discovered by Indian astrophysicists at **NCRA-TIFR**, Pune.
- **Green Hydrogen Plant for Railways:** India's first green hydrogen production plant for powering trains commissioned at **Jind, Haryana**.
- **First Indigenous Pilot Trainer Aircraft 'HANSA-3 NG' (29 Nov 2025):** Production version launched by **CSIR-National Aerospace Laboratories (CSIR-NAL)**, Bengaluru; features an all-composite airframe designed for civil (PPL/CPL) and military pilot training.
- **Agni-Prime Rail-Based Test (24 Sept 2025):** India successfully launched the new-generation surface-to-surface ballistic missile **Agni-Prime** (strike range: **1,000 to 2,000 km**; payload: 1,500–3,000 kg) from a specially designed **rail-based mobile launcher**, placing India among select nations with canisterised rail launch capability.
- **TASL Defence Facility in Morocco (23 Sept 2025):** Defence Minister Rajnath Singh and Moroccan counterpart inaugurated **Tata Advanced Systems Limited's (TASL)** defence manufacturing facility in **Berrechid, Morocco** — **India's first overseas private defence manufacturing plant**, producing the **Wheeled Armoured Platform (WhAP) 8x8** for the Royal Moroccan Army.
- **ICGS 'Adamya' Commissioned (19 Sept 2025):** First vessel of the **Adamya-class** (series of 8 Fast Patrol Vessels built by Goa Shipyard Limited) commissioned at **Paradip Port, Odisha**.
- **INS Arnala Commissioned (18 June 2025):** First of eight indigenously designed and built Anti-Submarine Warfare Shallow Water Crafts (ASW-SWC) commissioned at Visakhapatnam; built by **GRSE** in PPP with **L&T Shipyard**; named after historic Arnala Island off Vasai, Maharashtra (Chhatrapati Shivaji Maharaj legacy).
- **INS Nistar Commissioned (18 July 2025):** India's **first indigenously designed and built Diving Support Vessel (DSV)** commissioned at Visakhapatnam; built entirely by **Hindustan Shipyard Limited (HSL)** for deep-sea submarine rescue operations.

- **INS Tamal Commissioned (1 July 2025):** Commissioned at **Yantar Shipyard in Kaliningrad, Russia**; 8th stealth frigate under **Project 1135.6 (Tushil-class)**; officially marked as the **last foreign-built warship of the Indian Navy**.
- **Astra BVRAAM Test (11 July 2025):** DRDO and IAF flight-tested the Astra Beyond Visual Range Air-to-Air Missile equipped with an **indigenous active RF seeker** from a Su-30MKI fighter off Odisha, confirming strike capability over **100 km**.
- **Akash Prime High-Altitude Test (16 July 2025):** Tested by Army Air Defence and DRDO in high-altitude **Ladakh**; features an active RF seeker; seamlessly integrated with **'Akashteer'** (indigenous automated air defence control system).
- **Pralay Quasi-Ballistic Missile User Trials (28–29 July 2025):** Successful consecutive launches of surface-to-surface solid-fuel missile (range: **150–500 km**; payload: 350–700 kg) from Dr. APJ Abdul Kalam Island.
- **Agni-5 Launch (20 August 2025):** Strategic Forces Command (SFC) tested canister-based ICBM/IRBM Agni-5 (declared range **up to 5,000 km**) from Chandipur.
- **Rudrastra Hybrid Drone Test (11 June 2025):** Indian Army tested **Rudrastra**, a hybrid VTOL UAV developed by **Solar Defence & Aerospace Limited (SDAL)** at Pokhran (range: **170 km**, flight endurance: **90 minutes**).
- **Dhvani Hypersonic Missile:** DRDO advanced development of **Dhvani**, a next-generation **Hypersonic Glide Vehicle (HGV)** capable of speeds exceeding **Mach 5 (>7,400 km/h)** and an operational range of **6,000 to 10,000 km**.
- **Indian Space Industry E-Catalogue:** Launched as India's first digital repository mapping domestic space-sector capabilities and industrial supply chains during the **5th Indian Space Congress 2026** in New Delhi.
- **India's First Private EO Satellite Constellation (EO-PPP):** On 21 January 2026, **IN-SPACe** signed an agreement with the Pixxel-led consortium **"Allied Orbits"** (Pixxel Space India, Piersight Space, SatSure Analytics, Dhruva Space) to invest over **₹1,200 crore** across 5 years to launch **12 indigenous Earth Observation satellites** (panchromatic, multispectral, hyperspectral, and microwave SAR).

January 2026

LR-AShM hypersonic anti-ship missile | shown on Republic Day 2026

What happened On **26 January 2026**, during the **77th Republic Day** parade at **Kartavya Path**, DRDO showcased the **Long Range Anti-Ship Hypersonic Missile (LR-AShM)**. It is being developed mainly for the **Indian Navy**, including coastal battery use. Work is linked with DRDO's **RCI** lab in Hyderabad.

Core facts

- Full short name: **LR-AShM**.
- Maker: **DRDO**.
- User focus: **Indian Navy**.
- Reported range: about **1,500 km**.
- It is a **hypersonic glide** missile. It can hit both fixed and moving ship targets.
- Public briefings stress indigenous sensors and avionics.
- It uses a two-stage solid rocket boost, then glides at very high speed at low altitude, which makes early radar detection harder.

Captain Hansja Sharma — Rudra Armed Helicopter Milestone (26 Jan 2026)

- 27-year-old **Captain Hansja Sharma** created military history by becoming the **Indian Army's first woman pilot qualified to fly the Rudra armed attack helicopter**.

Solid Fuel Ducted Ramjet (SFDR) Test | 3 Feb 2026

- DRDO successfully flight-tested the indigenous **Solid Fuel Ducted Ramjet (SFDR)** technology at ITR Chandipur, enabling extended beyond-visual-range (BVR) air-to-air missile propulsion.

> **Confused pair**

BrahMos is a **supersonic cruise** missile. **Pralay** is a different tactical strike missile. **LR-AShM** is the hypersonic anti-ship story of January 2026.

K-4 missile test from INS Arighat

What happened India tested the **K-4** submarine-launched ballistic missile from **INS Arighat** in the Bay of Bengal (covered heavily in January defence current affairs).

Core facts

- Type: intermediate-range **SLBM** (missile fired from a submarine).
- Maker: **DRDO**.
- Reported range: more than **3,500 km**.
- Platform: **INS Arighat**, India's second nuclear-powered ballistic missile submarine in public discussion after Arihant.
- Why it matters: it strengthens the sea part of India's nuclear triad (land, air, and sea).

Pralay salvo test

What happened DRDO fired **two Pralay missiles** quickly from the **same launcher** near **Chandipur** in Odisha. This is called a **salvo** launch.

Core facts

- System: **Pralay**.
- Place: **Chandipur**, Odisha.
- Special point: two missiles from one launcher in quick succession.
- Role in simple words: precise tactical strike missile.

Other short defence notes from January coverage

- **Pinaka LRGR-120** was reported with about **120 km** range class testing from Chandipur.
- **Suryastra Universal Multi-Calibre Rocket Launcher:** In January 2026, the Indian Army fast-tracked a **₹293 crore** contract under emergency procurement powers with private manufacturer **NIBE Limited** (in collaboration with **Israel**) for **Suryastra**, India's first Made-in-India universal multi-calibre rocket launcher with strike ranges of **150 km and 300 km**, capable of also firing loitering munitions up to 100 km.
- **'Saksham' C-UAS Grid System:** Indian Army initiated induction of the indigenously developed **Saksham** (*Situational Awareness for Kinetic Soft and Hard Kill Assets Management*) Counter-Unmanned Aerial System grid, developed by **Bharat Electronics Limited (BEL)** to detect, track, and neutralise hostile quadcopters and swarm drones in border sectors using integrated soft-kill (electronic jamming) and hard-kill capabilities.
- Keep extra rocket or drone headlines only if a clear PIB fact appears later.

February 2026

Budget push for deep tech manufacturing | 1 February 2026

These are Budget ideas, but they often appear in Science questions too. Full scheme cards are in Schemes.

- **ISM 2.0** for semiconductors.
- **Biopharma SHAKTI** for advanced biological medicines.
- **Rare Earth Corridors** in Odisha, Kerala, Andhra Pradesh, and Tamil Nadu.
- **Three Chemical Parks**.
- **Container Manufacturing** scheme.
- **CCUS** (Carbon Capture, Utilization and Storage) with about **₹20,000 crore** over five years.
- Electronics Components Manufacturing outlay raised to **₹40,000 crore**.

IIT Kanpur maps the Sun's inside

What happened A research team at **IIT Kanpur** used about **30 years** of satellite data in a computer dynamo model. They mapped large magnetic fields inside the Sun. The work helps better understand sunspots, flares, and space weather that can disturb satellites and power grids.

Core facts

- Institute: **IIT Kanpur**.
- Method: long satellite record + 3D dynamo model.
- Use: better solar-cycle and space-weather understanding.

Greater Noida semiconductor story

February UP notes mentioned a semiconductor unit push in **Greater Noida (Gautam Buddh Nagar)** as North India's first such unit. Treat the "first" claim carefully until the plant status is fully clear. Keep the place fact under UP Special.

India-AI Impact Summit 2026 | New Delhi | 16–21 February 2026

What happened The Government of India hosted the **India-AI Impact Summit 2026** at **Bharat Mandapam, New Delhi** from **16 to 21 February 2026**. This was the **first time** the summit was hosted by a Global South nation.

Core framework: Three Sutras & Seven Chakras

- **Three Guiding Sutras:** 1. **People:** AI must serve humanity in all its diversity, respecting cultural identities and leaving no one behind. 2. **Planet:** Resource-efficient AI development accelerating climate resilience and green computing. 3. **Progress:** Equitable distribution of AI benefits; democratising datasets, compute, and foundational models for healthcare, education, agriculture, and governance.
- **Seven Action Chakras:** 1. *Human Capital:* Skilling and workforce transformation. 2. *Inclusion for Social Empowerment:* Preserving linguistic and cultural diversity. 3. *Safe and Trusted AI:* Safety evaluation, red-teaming, and auditing tools. 4. *Resilience, Innovation, and Efficiency:* Energy-efficient compute and algorithms. 5. *Science:* Accelerating scientific discovery and research. 6. *Democratizing AI Resources:* Public access to compute and datasets. 7. *AI for Economic Development & Social Good:* High-impact public sector deployments.
- **Official Logo:** The **Ashoka Chakra**, symbolizing ethical governance, constitutional justice, and democratic values guiding India's digital trajectory.
- **Global Launch:** Unveiled the **Global AI Impact Commons**, cataloguing 80+ real-world deployment stories across 30+ nations.

Bodhan AI & Bharat EduAI Stack | IIT Madras

What happened At the Bharat Bodhan AI Conclave 2026 at Bharat Mandapam (12–13 February 2026), Union Education Minister **Dharmendra Pradhan** launched the **Bodhan AI Centre of Excellence** and initiated the **Bharat EduAI Stack**.

Core facts - **Anchor Institution: IIT Madras** (established as the *IIT Madras Bodhan AI Foundation*). - **Core Architecture:** Conceived as an open-source **Sovereign Digital Public Infrastructure (DPI)** for education. - **Components:** Personalised generative AI student tutors, AI-assisted pedagogical support and lesson planning for teachers, automated institutional workflows, and learning analytics.

NIELIT Quantum & AI University Campus | Amaravati Quantum Valley

- On **20 February 2026**, the National Institute of Electronics and Information Technology (**NIELIT**, under MeitY) signed an MoU with the Government of Andhra Pradesh to establish India's first dedicated **Quantum and Artificial Intelligence University Campus** in **Amaravati** (*Amaravati Quantum Valley*).

New Telescopes in Ladakh & COSMOS-2 | Union Budget 2026–27 Push

The Union Budget 2026–27 announced dedicated capital support for 4 major ground-based astronomy facilities:

1. **National Large Solar Telescope (NLST):** - **Location: Merak**, near Pangong Lake, **Ladakh**. - **Specs: 2-metre-class** ground-based solar telescope observing in optical and near-infrared wavelengths. - **Objective:** Resolving the origin, dynamics, and violent eruptions of solar magnetic fields with high spatial resolution.
2. **National Large Optical-Infrared Telescope (NLOT):** - **Location: Hanle, Ladakh** (near the Indian Astronomical Observatory). - **Specs:** Flagship **13.7-metre segmented primary mirror** equipped with advanced adaptive optics, high-resolution spectrographs, and infrared imagers.
3. **Upgrade of Himalayan Chandra Telescope (HCT):** - **Location:** Hanle, Ladakh (altitude 4,500 m); remotely operated by the **Indian Institute of Astrophysics (IIA)** from CREST (Hosakote near Bengaluru). - **Upgrade:** Upgraded from the existing 2-metre mirror to a **3.7-metre segmented optical-infrared telescope** tailored for multi-messenger astronomy and transient cosmic events.
4. **COSMOS-2 Planetarium:** - **Location: Amaravati, Andhra Pradesh**; established and implemented by the **Indian Institute of Astrophysics (IIA)**.

Agni-III Intermediate-Range Ballistic Missile Test | 6 February 2026

What happened India successfully test-fired the indigenous intermediate-range ballistic missile **Agni-III** from the Integrated Test Range (ITR) at **Chandipur, Odisha**, under the Strategic Forces Command.

Core facts - **Type:** Indigenous, two-stage, solid-fuelled **Intermediate-Range Ballistic Missile (IRBM)**. - **Strike Range & Payload:** Operational strike range exceeding **3,000 km** with a **1.5-tonne** payload capacity. - **Agni Series Range Hierarchy:** - *Agni-1:* ~700 km - *Agni-2:* ~2,000 km - *Agni-3:* >3,000 km - *Agni-4:* ~4,000 km - *Agni-5:* >5,000 km (Intercontinental range; tested with MIRV in Mission Divyastra).

Launch of First Cadet Training Ship 'Krishna' | 16 February 2026

What happened The Indian Navy launched **'Krishna' (Yard 18003)**, the first of three indigenous **Cadet Training Ships (CTS)**, at **L&T Shipyard, Kattupalli (Chennai)**.

Core facts - **Vessel Name: Krishna** (first ship of the 3-vessel CTS order). - **Builder: Larsen & Toubro (L&T)** Shipbuilding, Kattupalli, Chennai. - **Dignitaries:** Launched by Smt. Anupama Chauhan in the presence of Chief of Defence Staff (**CDS**) **General Anil Chauhan**. - **Role:** Comprehensive sea training for officer cadets, including basic navigation, seamanship, and international naval diplomacy.

Additional Boeing P-8I Maritime Patrol Aircraft Procurement | 12 February 2026

- On **12 February 2026**, the Defence Acquisition Council (DAC) accorded Acceptance of Necessity (**AoN**) for the procurement of additional **Boeing P-8I Neptune** long-range maritime patrol and anti-submarine warfare (ASW) aircraft to counter increasing submarine activity in the Indian Ocean Region.

India AI Mission: GPU Scale & UDAAN Pitch Fest | February 2026

- **Compute Milestone:** Against an initial target of 10,000 GPUs, India expanded national public compute to **38,000 GPUs** and approved **570 Data Labs** across the country.
- **UDAAN Pitch Fest:** Launched during the India-AI Impact Summit to fund AI startups from Tier-2 and Tier-3 cities, prioritizing women founders and differently-abled changemakers.

Bharat Forecast System (BFS) Adopted by IMD | February 2026

What happened The India Meteorological Department (**IMD**) announced the adoption of the **Bharat Forecast System (BFS)**, India's first high-resolution indigenously engineered numerical weather model.

Core facts - **High-Resolution Spatial Grid:** Delivers weather predictions at a **$6\text{ km} \times 6\text{ km}$ spatial resolution**, replacing the legacy $12\text{ km} \times 12\text{ km}$ (144 sq km) model (research ongoing to scale down to 3 km and 1 km grids). - **Significance:** Radically enhances hyperlocal forecasting accuracy for severe weather events, extreme rainfall, tropical cyclones, and flash floods.

March 2026

Defence Acquisition Council clearances | 27 March 2026

What happened The **Defence Acquisition Council (DAC)**, chaired by Defence Minister **Rajnath Singh**, gave **Acceptance of Necessity (AoN)** for big purchases worth about **₹2.38 lakh crore**. AoN is an early formal yes that starts the full buying process.

Important items cleared

- More **S-400** long-range surface-to-air missile systems from Russia for the Air Force.
- **Medium Transport Aircraft** to slowly replace older transport planes such as **An-32** and help with **Il-76** class airlift needs. Indian companies will build them with a foreign technology partner.
- **Remotely Piloted Strike Aircraft** linked with DRDO's **Ghatak** unmanned combat aircraft work. Value in reports is near **₹40,000 crore** for this line.
- More indigenous **Dhanush** 155 mm howitzer guns for the Army.
- Other Army items named in briefings: air defence tracked systems, better tank ammunition, high-capacity radio relay, and runway-independent aerial surveillance.
- Coast Guard also got approvals such as heavy-duty air cushion vehicles in coverage.

Year context

- In FY 2025–26, DAC AoN total was described as about **₹6.73 lakh crore** across 55 proposals — called a record year in official messaging.

NavIC / IRNSS-1F atomic clock failure | 13 March 2026

What happened ISRO said the atomic clock on navigation satellite **IRNSS-1F** stopped working on **13 March 2026**. Without a working atomic clock, the satellite cannot give proper positioning signals.

Core facts

- System name: **NavIC** (also called **IRNSS** — Indian Regional Navigation Satellite System).
- Satellite: **IRNSS-1F**, launched in **March 2016**; it had completed about **10 years** of design life.
- After this failure, the number of fully useful positioning satellites fell below the minimum **four** needed for reliable 3D fixes.
- The satellite can still be used for some one-way broadcast messaging.
- Newer **NVS** satellites carry improved Indian clocks. Replacement launches are planned later.

> **Confused pair**

NavIC is India's regional system. It is not the same as GPS (USA), GLONASS (Russia), Galileo (EU), or BeiDou (China).

Air-droppable container test | DRDO and Indian Navy

What happened DRDO and the Navy tested an indigenous **air-droppable container** from a **P-8I** aircraft off the **Goa** coast.

Core facts

- It can drop supplies of about **150 kg** to ships far at sea.
- Useful when a ship needs urgent stores without coming to harbour.

Project 'Vayu Baan' | March 2026

- The Indian Air Force (**IAF**) launched **Project Vayu Baan** to develop India's first indigenous **helicopter-launched drone system**, enabling standoff surveillance and tactical strikes.

Haven-1 — World's First Commercial Space Station

- Aerospace company **Vast** announced timelines for **Haven-1**, scheduled to become the world's first commercial, privately operated space station in low Earth orbit.

CALM-Brain Repository | Bengaluru

- India's first digital repository of comprehensive clinical and neuro-imaging data on major psychiatric disorders, named **CALM-Brain**, was launched in **Bengaluru**.

Bharat Electricity Summit 2026 & Historic Clean Energy Milestones | March 2026

What happened The **Bharat Electricity Summit 2026** was held from **19 to 22 March 2026** at **Yashobhoomi, New Delhi**, themed *"Electrifying Growth. Empowering Sustainability. Connecting Globally."*

Core facts & Official National Power Data - **Strategic Blueprints Released:** - *National Resource Adequacy Plan*. - *Transmission Plan for Integration of over 900 GW Non-Fossil Fuel Capacity by 2035–36*. - **Historic Milestones (as of 31 March 2026):** - **Total Installed Power Generation Capacity: 5,32,740 MW (532.74 GW)**. - **Total Non-Fossil Fuel Capacity: 2,83,468 MW (283.47 GW)** — crossing **53.2% of total installed power capacity**, achieving India's COP26 Panchamrit pledge of >50% non-fossil power 4 years ahead of 2030! - **Single-Year Record Additions (FY 2025–26):** Non-fossil addition reached an all-time high of **55.29 GW** (comprising **44.61 GW solar** and **6.05 GW wind**). - **Latest Growth (as of 31 May 2026):** - Total non-fossil capacity reached **2,91,525.08 MW (291.53 GW)**. - Cumulative solar capacity reached **157.05 GW** (largest renewable segment).

Major Naval Deliveries & Inductions | Late March 2026

1. INS Dunagiri (Yard 3023) — Project 17A Stealth Frigate: - Delivered to the Indian Navy on **30 March 2026** by **Garden Reach Shipbuilders & Engineers (GRSE), Kolkata**. - Fifth ship of the Nilgiri-class guided-missile stealth frigates (second built by GRSE); preserves the naval heritage of the erstwhile Leander-class frigate that served from 1977 to 2010.

2. Anti-Submarine Warfare Shallow Water Crafts (ASW SWC): - **AGRAY:** Fourth of eight ASW-SWC built by GRSE Kolkata, delivered on **30 March 2026**; at 77 metres long, AGRAY and its sister ships are the **largest Indian Naval warships propelled by waterjets**! - **Malwan:** Delivered on **31 March 2026** by **Cochin Shipyard Limited (CSL)** under the Mahe-class ASW SWC line; named after historic coastal Malwan in Sindhudurg, Maharashtra (maritime legacy of Chhatrapati Shivaji Maharaj).

3. INS Sanshodhak (SVL): - Fourth and final **Survey Vessel (Large)** delivered by GRSE Kolkata for deep-water and port hydrographic surveying.

April 2026

Prototype Fast Breeder Reactor criticality | Kalpakkam | 6 April 2026

What happened India's **500 MWe Prototype Fast Breeder Reactor (PFBR)** at **Kalpakkam, Tamil Nadu** attained **first criticality** on **6 April 2026**, initiating a self-sustaining nuclear chain reaction.

Core facts - **Capacity & Builder: 500 MWe** (megawatt electrical); constructed by **BHAVINI** (Bharatiya Nabhikiya Vidyut Nigam Limited) at the Kalpakkam Nuclear Complex. - **Global Milestone:** Once fully operational, India will become only the **second country in the world after Russia** to operate a commercial fast breeder reactor! - **Core Fuel & Coolant:** Uranium–Plutonium **Mixed Oxide (MOX)** fuel; cooled by liquid **sodium**. - **India's Three-Stage Nuclear Programme (Dr. Homi Bhabha):** - **Stage 1 — PHWRs:** Pressurised Heavy Water Reactors using **natural uranium**; spent fuel yields **Plutonium-239**. - **Stage 2 — FBRs (PFBR Milestone):** Fast Breeder Reactors fueled by Plutonium; breeds **Uranium-233** from surrounding **Thorium-232 blankets** while generating more fissile fuel than consumed. - **Stage 3 — Thorium Reactors:** Harnesses India's vast domestic thorium reserves using bred Uranium-233 for indefinite energy self-reliance.

Artemis II Mission | NASA | 1 April 2026

What happened NASA launched the historic **Artemis II** crewed lunar flyby mission from the Kennedy Space Center on **1 April 2026**, sending humans beyond low Earth orbit toward the Moon for the first time in **53 years** (since Apollo 17 in December 1972).

Core facts - **Type:** Crewed lunar flyby demonstration mission (slingshot around the Moon, not a landing). - **Launcher & Spacecraft:** Launched atop the **Space Launch System (SLS)** rocket; Orion spacecraft named **'Integrity'** by the crew. - **Engineering Validation:** Verified life support systems, deep-space navigation, manual flight control, and the critical **Avcoat** base heat shield during high-speed atmospheric re-entry. - **Next Step:** NASA lists **Artemis III** for 2027 as a low Earth orbit demo to test commercial lunar human landing systems from **SpaceX** and **Blue Origin**.

SMOPS-2026 (Spacecraft Mission Operations) | Bengaluru | 8–10 April 2026

What happened The 2nd edition of the **International Conference on Spacecraft Mission Operations (SMOPS-2026)** was held in **Bengaluru** from **8 to 10 April 2026**.

Core facts - **Theme:** *"Innovative Operations for Smart and Sustainable Space Mission Management – Next Generation"*. - **Organisers:** Jointly organised by **ISRO**, the **Astronautical Society of India (ASI)**, and the **International Academy of Astronautics (IAA)**. - **Inauguration:** Inaugurated by Shri **A. S. Kiran Kumar**, former Chairman of ISRO. - **Scope:** Mission autonomy, automated constellation management, deep-space communication, and space robotics.

Chandrayaan-5 (LUPEX Mission) | April 2026

- ISRO and JAXA (Japan Aerospace Exploration Agency) began evaluating launch infrastructure in **Japan** for the joint **Lunar Polar Exploration (LUPEX)** mission, officially designated as **Chandrayaan-5**.

Cochin Airport — Hydrogen Buses

- **Cochin International Airport** became the first airport in India to operationalise zero-emission **hydrogen-powered buses** for internal airside passenger transit.

> **Easy wrong options**

Calling PFBR a pressurised heavy water reactor. Mixing Kalpakkam with Kudankulam. Saying it already runs fully on thorium fuel today.

INS Aridhaman and INS Taragiri | Visakhapatnam | 3 April 2026

What happened Defence Minister **Rajnath Singh** commissioned naval platforms at **Visakhapatnam** on **3 April 2026**. Coverage confirmed induction of India's third indigenous nuclear-powered ballistic missile submarine and the fourth Project 17A stealth frigate.

INS Aridhaman

- Type: nuclear-powered ballistic missile submarine (**SSBN**).
- Sequence: third after **INS Arihant** and **INS Arighaat**.
- Project family: Advanced Technology Vessel (**ATV**) / Arihant-class line.
- Stronger second-strike capability is the strategic point.

INS Taragiri

- Fourth ship of **Project 17A** (**Nilgiri-class**) guided-missile stealth frigates.
- Displacement: about **6,670 tonnes**.
- Built by **Mazagon Dock Shipbuilders Limited**.
- Designed by the Navy's **Warship Design Bureau**.

Also at Vizag

- Foundation stone for a **Large Cavitation Tunnel** facility at DRDO's Naval Science & Technological Laboratory.

NASM-SR maiden salvo launch | off Odisha coast

What happened DRDO and the Navy carried out the first **salvo launch** of two **Naval Anti-Ship Missile – Short Range (NASM-SR)** missiles from a helicopter over the Bay of Bengal off **Odisha**.

Core facts

- NASM-SR is India's **first indigenous air-launched anti-ship missile**.
- Range noted: about **55 km**.
- Features: low-level sea-skimming, fire-and-forget, all-weather use.
- Test range support: **ITR Chandipur**.
- Salvo means two missiles launched quickly from the **same** helicopter.

INS Malwan delivery (soft navy fact)

- Second of eight **Anti-Submarine Warfare Shallow Water Craft** from **Cochin Shipyard**.
- Named after the coastal town of **Malwan** in Maharashtra.

May 2026

Advanced Agni MIRV flight trial | 8 May 2026 | Odisha

What happened India successfully flight-tested an advanced **Agni** missile with **MIRV** technology from **Dr APJ Abdul Kalam Island** off Odisha. MIRV means one missile can carry several warheads that hit different targets.

Core facts

- Date: **8 May 2026**.
- Place: **Dr APJ Abdul Kalam Island** / Odisha coast.
- Technology: **MIRV** (Multiple Independently Targetable Re-entry Vehicle).
- Linked in coverage with the advanced **Agni-5** line, earlier called **Mission Divyastra** (first known MIRV test in **March 2024**).
- Agni-5 declared range memory: over **5,000 km**.
- Conducted with DRDO, Army, and industry support.

> **Easy memory**

One missile → many warheads → many targets. That is MIRV.

Agni-1 test | 22 May 2026

What happened India successfully test-launched the short-range ballistic missile **Agni-1** from the Integrated Test Range at **Chandipur / Balasore, Odisha**, under the **Strategic Forces Command**.

Scramjet Combustor Test | DRDL Hyderabad | May 2026

- DRDL Hyderabad conducted a successful long-duration ground test of an indigenous **scramjet combustor** exceeding **1,200 seconds** to validate thermal endurance and supersonic combustion for future hypersonic platforms.

India's 1st Indigenous Glide Weapon System — TARA | 7 May 2026

What happened DRDO and the Indian Air Force (IAF) successfully conducted the maiden flight trial of **TARA** (Tactical Advanced Range Augmentation) off the coast of Odisha on **7 May 2026**.

Core facts - **System Name: TARA** (**Tactical Advanced Range Augmentation**). - **Type:** India's **first indigenous glide weapon system**. - **Developer:** Designed and developed by **Research Centre Imarat (RCI), Hyderabad**, along with other DRDO laboratories. - **Function:** A modular range-extension and precision-guidance kit that converts conventional unguided gravity aerial bombs/warheads into precision-guided standoff glide weapons. - **Combat Advantage:** Drastically elevates the strike value of existing low-cost aerial munitions, enabling IAF aircraft to strike heavily defended targets from outside enemy air defence envelopes.

ICGS 'Achal' Commissioned into Indian Coast Guard | 9 May 2026

What happened Indian Coast Guard Ship (**ICGS**) **Achal** was commissioned into service on **9 May 2026** at **Goa Shipyard Limited (GSL)**.

Core facts - **Class:** Latest vessel in the new-generation **Adamya-class** Fast Patrol Vessel (FPV) series. - **Builder:** Designed and built indigenously by **Goa Shipyard Limited (GSL)**. - **Etymology:** Named **"Achal"**, a Sanskrit term signifying **"firm / immovable"**. - **Role:** High-speed coastal patrol, anti-smuggling, EEZ maritime surveillance, and fisheries protection.

Oreshnik Hypersonic Intermediate-Range Ballistic Missile | 24 May 2026

- On **24 May 2026**, Russia fired the **Oreshnik** (Russian for *"Hazel Tree"*, Ukrainian reporting codename *"Kedr"* / Cedar) intermediate-range ballistic missile (IRBM) during an attack on Kyiv.
- Characterised by multiple hypersonic warheads descending in steep, fiery streaks at speeds virtually impossible to intercept by conventional Western missile defence shields.

Exercise Pragati 2026 | Umroi, Meghalaya

What happened The maiden multilateral exercise **Pragati 2026** brought armies of **13** countries to **Umroi, Meghalaya**. Focus: counter-insurgency drills and interoperability in the Indian Ocean region.

Core facts

- Host place: **Umroi**, Meghalaya.
- About **400** soldiers in coverage, including women soldiers.
- Partner countries named: Bhutan, Cambodia, Indonesia, Laos, Malaysia, Maldives, Myanmar, Nepal, Philippines, Seychelles, Sri Lanka, Vietnam, and India.

Exercise CINBAX 2026

- Edition: **2nd**.
- Countries: **India** and **Cambodia**.
- Place in coverage: **Kampong Speu**, Cambodia.

First woman Cat-A flying instructor | IAF

- Name in coverage: Squadron Leader **Saanya**.
- Qualification: Category-A Qualified Flying Instructor (**Cat-A QFI**).

VISTA Stratospheric Balloon | May 2026

What happened India's first indigenous **stratospheric super-pressure balloon (SPB)**, named **VISTA**, was successfully developed and launched.

Core facts - **Developer: Red Balloon Aerospace** (private deep-tech space startup). - **Launch Site: Vijayawada, Andhra Pradesh**. - **Capability:** Long-duration high-altitude atmospheric observations, near-space research, and high-altitude surveillance.

SMILE Space Mission | May 2026

- **Agencies:** Joint mission of the **Chinese Academy of Sciences (CAS)** and the **European Space Agency (ESA)**.
- **Full Form: Solar wind Magnetosphere Ionosphere Link Explorer**.
- **Objective:** Deep study of solar wind interactions with Earth's magnetosphere and cosmic weather.

India's First Orbital Data Centre Satellite — Pathfinder | May 2026

What happened On **4 May 2026**, private space-tech firm **Pixxel** and Indian AI startup **Sarvam** announced a strategic partnership to build **Pathfinder**, India's first orbital data centre satellite.

Core facts - **Satellite Class & Launch:** Planned as a **200 kg-class satellite** scheduled for launch in **Q4 2026**. - **Manufacturing:** Built at Pixxel's **Gigapixxel** manufacturing facility (engineered to scale satellite production to 100 units). - **Pixxel's Role:** Designs, builds, launches, and operates the satellite platform; integrates hyperspectral imaging sensors. - **Sarvam's Role:** Delivers the full-stack sovereign AI platform running directly on the satellite's GPU layer for real-time onboard training and edge inference in orbit. - **Strategic Impact:** Eliminates the latency of sending massive raw imagery to terrestrial ground stations; guarantees national AI sovereignty in space edge computing.

Mission Drishti: World's First OptoSAR Satellite | May 2026

What happened Indian private space startup **GalaxEye** developed **Mission Drishti**, hailed by the Prime Minister's Office as the **world's first OptoSAR satellite** and the **largest privately built satellite in India**.

Core facts - **OptoSAR Innovation:** Overcomes the trade-off of Earth observation by integrating both **Multispectral Imaging (MSI)** optical sensors and **Synthetic Aperture Radar (SAR)** sensors on a single satellite bus. - **Advantage:** Optical sensors provide high-resolution, intuitive visual clarity but are blocked by clouds, haze, rain, and night; SAR penetrates clouds and darkness but lacks optical intuition. OptoSAR delivers synchronized, all-weather, analysis-ready imagery.

Chandrayaan-3 Hop Experiment & Regolith Discoveries | May 2026

What happened On **18 May 2026**, ISRO scientists published comprehensive peer-reviewed findings from the historic **Hop experiment** conducted by the **Vikram Lander** on the lunar South Pole.

Core facts - **The Maneuver:** Before shutting down, Vikram reignited its liquid thrusters, performed a controlled **50 cm hop**, and relocated; its **ChaSTE** (Chandra's Surface Thermophysical Experiment) thermal probe re-penetrated virgin regolith at the new site. - **Major Scientific Insights:** 1. **Layered Stratigraphy:** The lunar regolith within the top **6.5 cm** is not homogenous dust; it exhibits distinct stratification with sharp physical and thermal boundaries. 2. **Geotechnical Compaction:** Soil density and cohesion increase drastically with depth — transitioning from loose dust to densely packed, sticky regolith. 3. **Twilight Thermal Shadowing:** Micro-depressions and local craters cast sharp thermal shadows, causing surface temperatures to plummet much faster than predicted by global lunar climate models. 4. **Plume Erosion:** The engine plume stripped away the upper **3 cm** of regolith, exposing and slightly compacting the deeper cohesive substratum.

June 2026

Indian Navy platforms commissioned | GRSE line

What happened Coverage highlighted commissioning of indigenous Navy platforms designed by the Warship Design Bureau and built by **Garden Reach Shipbuilders and Engineers (GRSE)**.

Core facts

- **INS Dunagiri** — Project **17A** stealth frigate (BrahMos and medium-range air defence in coverage).
- **INS Sanshodhak** — survey / hydrographic vessel.
- **INS Agray** — anti-submarine warfare craft.

Indian Coast Guard additions

- **ICGS Akshay** — new-generation Fast Patrol Vessel, built by **Goa Shipyard Limited**, commissioned in Goa.
- First indigenous **Air Cushion Vehicle** for the Coast Guard: **H-561**, first of six being built by Chowgule and Company.

Rudram-II Flight-Test | 2 June 2026

What happened The Defence Research and Development Organisation (**DRDO**) and the **Indian Air Force (IAF)** successfully flight-tested the indigenous **Rudram-II** missile from a Su-30MKI fighter aircraft off the Odisha coast.

Core facts - **Type:** Indigenous **Air-to-Surface** anti-radiation and precision-strike missile. - **Developer:** DRDO. - **Role:** Neutralising enemy radar installations, communication nodes, and surface-to-air missile (SAM) sites.

NASA MAVEN Mission Concluded | June 2026

- NASA announced the conclusion of the long-running **MAVEN** (*Mars Atmosphere and Volatile EvolutioN*) orbital science mission exploring the history of Mars' atmospheric and water loss.

Soft space–book hybrid

Group Captain **Shubhanshu Shukla**'s memoir coverage recalled him as the first Indian citizen on the **International Space Station** (Axiom-4). Book card → Books.

Bharat Innovates 2026 | Nice, France | 14–16 June 2026

What happened The **Bharat Innovates 2026** deep-tech summit concluded in **Nice, France** (14–16 June 2026), marking a major milestone in global institutional partnerships and commercial venture backing for Indian indigenous science.

Core facts - **Participants:** Showcased **120 curated Indian DeepTech startups** and **45 advanced technology projects** from 15 premier Indian institutions (over 60% incubated at IITs and IISc). - **Major Commercial Partnerships:** - **Agnikul Cosmos** signed agreements with **ICEYE** (Finland) and **Safran** (France). - **Detect Technologies** partnered with **TotalEnergies**. - **TuTr Hyperloop** collaborated with **thyssenkrupp**. - **iElectron Technologies** partnered with **ALTEN**. - Confirmed strategic angel and institutional investment into **The ePlane Company**.

India's First Advanced 3D Semiconductor Packaging Unit | Bhubaneswar, Odisha

What happened The foundation stone for India's first advanced **3D semiconductor chip packaging facility** based on glass substrates was laid in **Bhubaneswar, Odisha**.

Core facts - **Promoter & Project:** Promoted by **3D Glass Solutions** under the *Heterogeneous Integration Packaging Solutions* project. - **Facility Type:** Greenfield, vertically integrated advanced packaging and embedded glass-substrate ATMP (Assembly, Testing, Marking, and Packaging) unit. - **Financial Outlay:** Total project investment of **₹1,943.53 crore** approved under Phase 1 of the **India Semiconductor Mission (ISM)**. - **Strategic Milestone:** Employs glass instead of traditional organic substrates for 3D heterogenous chip stacking, drastically improving heat dissipation and high-frequency performance; makes **Odisha the first State** to host both an approved compound semiconductor fab and a 3D glass packaging plant.

Operation Mule Hunt 1.0 | Gujarat Cyber Police | June 2026

- **What happened:** Gujarat Police and the Cyber Centre of Excellence (CCOE) carried out **Operation Mule Hunt 1.0**, dismantling statewide cyber financial crime syndicates using mule bank accounts.
- **Key Outcomes:** Uncovered cyber fraud worth **₹2,289 crore**, clamped down on **913 mule bank accounts**, registered 565 FIRs, and arrested 638 accused persons.

India Assumes Chairmanship of Common Criteria Development Board (CCDB)

- **Appointment:** India was unanimously nominated as Chair of the **Common Criteria Development Board (CCDB)** for a two-year tenure from **April 2026 to April 2028**.
- **Context:** Confirmed during the meeting of the Common Criteria Recognition Arrangement (CCRA) in **Tokyo, Japan**.
- **Role:** CCRA is the premier global standard ensuring mutual recognition of IT security certificates, eliminating repeated certification hurdles for Indian secure IT and cyber products across 30+ member nations.

July 2026

Project Kusha maiden flight test | 23 July 2026 | Odisha

What happened DRDO conducted the maiden flight test of **Kusha**, India's indigenous long-range surface-to-air missile (also called the Extended Range Air Defence System / earlier **XRSAM**), from **Dr APJ Abdul Kalam Island** off Odisha.

Core facts

- Date: **23 July 2026**.
- Place: **Dr APJ Abdul Kalam Island** / Odisha coast.
- Type: long-range **surface-to-air** missile.
- Lead user service: **Indian Air Force** (Navy shipborne version expected later).
- Development clearance memory: CCS outlay about **₹21,700 crore** (2022 window).
- Fits inside the wider **Mission Sudarshan Chakra** air-defence shield idea (target horizon noted around **2035**).
- Induction window noted: about **2028–2030**.

> **Layer memory**

Medium tier often linked with **Akash-NG**. Long-range indigenous tier: **Kusha**. Imported long-range memory: **S-400**.

Skyroot Vikram-1 | Mission Aagaman | 18 July 2026

What happened Skyroot Aerospace successfully launched **Vikram-1**, called India's first privately developed orbital-class rocket, from **SDSC SHAR, Sriharikota**. The flight was named **Mission Aagaman**.

Core facts

- Company: **Skyroot Aerospace**.
- Rocket: **Vikram-1**.
- Mission name: **Aagaman**.
- Place: **Sriharikota**.
- Meaning: private Indian orbital launch capability arrives.

Navy / Coast Guard July platforms

- **INS Mahendragiri** — sixth **Project 17A** stealth frigate, Eastern Fleet, Visakhapatnam.
- **Machilipatnam (BY 529)** — seventh of eight ASW Shallow Water Craft from Cochin Shipyard.
- **Operation Southern Readiness 26-2** — multinational training at Southern Naval Command, **Kochi**, with Combined Maritime Forces / CTF 154.

August 2026

National Space Day | 23 August 2026

What happened India marked the **3rd National Space Day** (remembering **Chandrayaan-3**). Theme in coverage: **Towards Viksit Bharat: Innovating, Collaborating and Aspiring to Global Space Leadership**. Events included Ahmedabad / SAC coverage.

Core facts

- Date every year: **23 August**.
- ISRO plan noted for the financial year: about **8** launch-vehicle missions and **12** satellite missions.
- First uncrewed **Gaganyaan** test flight expected within about four months in ISRO messaging.
- Soft policy note: ISRO to step back from routine launch-vehicle manufacturing and hand more work to private / PSU partners, while focusing on research (IN-SPACe messaging).

AK-203 'Sher' | Amethi, Uttar Pradesh

What happened The first **fully indigenous** AK-203 assault rifle, named **Sher**, was manufactured and test-fired at the **Korwa** facility in **Amethi, Uttar Pradesh**.

Core facts

- Name: **Sher**.
- Plant: **Korwa**, Amethi (UP).
- Maker: **Indo-Russian Rifles Private Limited (IRRPL)**.
- Claim in coverage: **100%** Indian components and materials.

- Aim: replace ageing **INSAS** rifles after Army trials.

INS Nipun and India–USA EOD exercise

- **INS Nipun** — second **Nistar-class** Diving Support Vessel, built by **Hindustan Shipyard Limited**, Visakhapatnam.
- **EOD Exercise 2026** (8th edition) — India and USA explosive ordnance disposal drills at Southern Naval Command, **Kochi**.

Strategic Doctrines & Nuclear Roadmap (Ghatna Chakra Eye Drishti 2026)

India's First National Counter-Terrorism Policy & Strategy — PRAHAAR | 23 February 2026

What happened The Ministry of Home Affairs (MHA) officially released India's first unified, formal counter-terrorism doctrine named **PRAHAAR**.

Core facts - **Doctrine Framework**: Establishes an integrated, multi-agency operational blueprint combating terrorism, radicalisation, cross-border terror funding, automated cyber warfare, weaponized drones, and **CBRNED threats** (Chemical, Biological, Radiological, Nuclear, Explosive, and Digital). - **Core Principle**: Expressly affirms that India rejects any conceptual link between terrorism and specific religions, ethnicities, or civilisations.

Defence Forces Vision 2047 | Released 10 March 2026

What happened Raksha Mantri Rajnath Singh released the apex long-term defence blueprint titled *"Defence Forces Vision 2047: A Roadmap for a Future-Ready Indian Military"* at South Block, New Delhi.

Core facts - **Formulated By**: **Headquarters Integrated Defence Staff (HQ IDS)**. - **Three-Phase Calibrated Transformation Roadmap**: 1. **Era of Transition (till 2030)**: Jointness, theaterisation, structural restructuring, rapid induction of indigenous drones, AI, and border surveillance networks. 2. **Era of Consolidation (2030–2040)**: Operationalizing layered anti-missile and integrated air defence grids, offensive cyber capabilities, and sovereign military space architecture. 3. **Era of Excellence (2040–2047)**: Achieving an entirely self-reliant, all-domain expeditionary military force by the centenary of Independence.

SHANTI Act, 2025 & Long-Term Nuclear Power Roadmap

What happened The **SHANTI Act, 2025** (received Presidential assent in December 2025) overhauls India's civilian nuclear legislative framework to dramatically ramp up clean base-load energy.

Core facts - **National Nuclear Target**: Expanding nuclear power capacity to **100 GW by 2047** (current installed capacity stands at **8.78 GW** or ~3.1% of national power generation; slated to reach 22.38 GW by 2031–32). - **Private Sector Entry**: Legally permits private companies to participate in nuclear plant construction, routine operations, equipment manufacturing, and small modular reactor (SMR) development. - **Strategic Sovereign Monopoly**: The Central Government strictly retains exclusive control over sensitive fuel-cycle operations, uranium enrichment beyond notified thresholds, heavy water production, and spent-fuel reprocessing/nuclear waste management. - **Dispute Redressal Bodies**: Establishes the **Nuclear Damage Claims Commission** and the **Atomic Energy Redressal Advisory Council**.

Defence Procurement Manual 2025 (DPM 2025)

- **Approval & Date**: Approved on **23 October 2025** by Defence Minister Rajnath Singh; came into effect on **1 November 2025**.
- **Predecessor**: Completely supersedes the outdated **DPM 2009**.
- **Crucial Prelims Distinction**: DPM 2025 strictly governs **Revenue Procurement** (routine maintenance, spare parts, rations, clothing, fuel, and minor equipment totaling ~**₹1 lakh crore annually**). It does **NOT** govern capital acquisitions (which are governed by the Defence Acquisition Procedure - DAP 2020).
- **Core Principles**: Enhances operational jointness among the Army, Navy, and Air Force, slashes procurement timelines, and mandates preferential sourcing for domestic MSMEs and startups under Atmanirbhar Bharat.

Indian Maritime Doctrine 2025

- **Release**: Released on **2 December 2025** by the Chief of the Naval Staff ahead of Navy Day (celebrated on **4 December 2025**).
- **Central Theme**: ***"Combat Ready, Cohesive, Credible, and Aatmanirbhar Force"***, articulating the Navy's operational readiness to secure vital Sea Lines of Communication (SLOCs) and protect India's maritime sovereignty in the Indo-Pacific.

India–UAE 8-Exaflop Supercomputing Cluster

- **Agreement**: Term sheet finalized between India's **Centre for Development of Advanced Computing (C-DAC)** and UAE's AI major **G42** during the May 2026 bilateral summit.
- **Scale**: Establishes a massive **8-exaflop supercomputing cluster** dedicated to training large language models (LLMs), climate modeling, and sovereign AI development across the Global South.

BSNL Nationwide VoWiFi (Wi-Fi Calling) Rollout

- **Launch Date**: **1 January 2026** (New Year 2026).
- **Technology**: Deployed **Voice over Wi-Fi (VoWiFi)** nationwide across all telecom circles, enabling mobile users to make and receive crystal-clear voice and video calls over Wi-Fi networks in cellular "shadow zones" (basements, remote locations, high-rises) without extra app installations.

Cameco–Department of Atomic Energy (DAE) Uranium Pact

- **Value**: **CAD 2.6 billion agreement** signed during Canadian PM Mark Carney's visit (Feb–March 2026).
- **Scope**: Canadian uranium giant **Cameco** guarantees long-term supply of natural uranium concentrates to the **Department of Atomic Energy (DAE)** to feed India's expanding fleet of Pressurised Heavy Water Reactors (PHWRs).

IN Operation Sindhu (West Asia Strategic Evacuation)

- **Mission**: Evacuation operation launched by the **Government of India** in 2026 to extract Indian citizens stranded across West Asian conflict zones (amid US–Israel–Iran military escalations).
- **Air & Maritime Dimensions**: Coordinated by the Ministry of External Affairs (MEA), Indian Air Force (IAF C-17 Globemaster III transport aircraft), and Indian Navy warships deployed on anti-piracy / maritime patrol in the Persian Gulf and Arabian Sea.

SIPRI 2026 Nuclear & Strategic Metrics

- **Nuclear Warheads Stockpile**: India's estimated nuclear stockpile expanded to **190 warheads** (surpassing Pakistan's 170 warheads). Global total: 12,187 warheads (Russia 5,420; USA 5,042; China 620; France 370; UK 225).
- **Global Military Spending**: India ranked **5th globally** with an annual defence budget of **$92.1 billion** (an 8.9% expansion over 2024; trailing US $954B, China $336B, Russia $190B, Germany $114B).
- **Arms Imports Share**: India is the **world's 2nd largest arms importer** accounting for **8.2% of global arms imports** in 2021–2025 (surpassed only by war-torn Ukraine at 9.7%).

IN Operation Sagar Bandhu (HADR Mission — Sri Lanka)

- **Context**: Disaster relief operation launched by India in late November 2025 following widespread destruction caused by **Cyclone Ditwah** in Sri Lanka (*Sagar Bandhu* = "Ocean Friend").
- **Naval & Air Assets**: Indian Navy deployed aircraft carrier **INS Vikrant**, stealth frigate **INS Udaygiri**, and offshore patrol vessel **INS Sukanya**, alongside Indian Air Force **C-130J Super Hercules** transport aircraft with emergency stores, field hospital equipment, and humanitarian supplies.

INTERPOL Silver Notice (Illicit Financial Assets)

- **Launch Date**: Formally introduced in **January 2025** as the 8th official color-coded alert of INTERPOL.
- **Mandate**: Designed specifically to **identify, track, freeze, and seize illicit criminal assets and money-laundering proceeds** across international jurisdictions, strengthening global asset recovery.

CR450 Next-Gen High-Speed Train

- **Record**: China unveiled the **CR450 bullet train**, reaching an operational test speed of **453 km/h**, setting the benchmark for the world's fastest conventional passenger rail technology.

BharatGen: India's First AI Multimodal Large Language Model (MLLM)

- **Launch Date & Forum**: Launched on **2 June 2025** by Union Minister of State (IC) for Science & Technology at the BharatGen Summit 2025.
- **National Breakthrough**: India's first indigenously developed **Multimodal Large Language Models (MLLMs)**.
- **Linguistic Breadth**: Purpose-built to process and generate content across **22 Indian languages**.

- **Multimodal Capabilities**: Processes heterogeneous data formats simultaneously — text, images, audio, and video — unlike older unimodal models.
- **Mission Framework**: Developed under the **National Mission on Interdisciplinary Cyber-Physical Systems (NM-ICPS)**.
- **Executing Agency**: **TIH Foundation for IoT and IoE at IIT Bombay** (Technology Innovation Hub).

Kalaa Setu & BhashaSetu — Real-Time Language Tech for India

- **Initiative**: Launched by the **Ministry of Information & Broadcasting (MIB)** through its **WaveX Startup Accelerator Platform**.
- **WaveX Platform**: Dedicated startup accelerator launched under MIB's flagship **WAVES initiative** in March 2025 to nurture innovation across media, entertainment, and language technologies.
- **BhashaSetu Challenge**: Real-time language translation challenge launched under WaveX on **30 June 2025**.
- **Kalaa Setu Challenge**: Designed to advance real-time AI-powered multilingual speech and translation tools for inclusive, last-mile citizen communication.

Param Rudra Supercomputer Inaugurated at IIT Patna (Bihar's First Supercomputer)

- **Milestone**: Patna joined India's high-performance computing map with the inauguration of **Param Rudra**, the **first supercomputer in Bihar**.
- **Mission**: Installed under the **National Supercomputing Mission (NSM)**.
- **Collaboration**: Institutionalized through an MoU between **IIT Patna** and the **Centre for Development of Advanced Computing (C-DAC)**.
- **Computing Power**: Operates at **838 Teraflops (TF)**, utilizing predominantly indigenous components and running on an Indian-developed system software stack created by C-DAC.

India's First LNG-Diesel Dual-Fuel DEMU Train

- **Operational Debut**: Indian Railways launched the country's first **LNG-diesel dual-fuel DEMU train** at **Sabarmati, Ahmedabad**.
- **Technical Retrofit**: Driving power cars of the **1,400 HP Diesel Electric Multiple Unit (DEMU)** were retrofitted with a dual-fuel injection system.
- **Fuel Economics & Emissions**: Utilizes Liquefied Natural Gas (LNG) to replace approximately **40% of traditional diesel consumption**, substantially curbing particulate emissions and operating expenditure.

National Shipbuilding & Heavy Industries Park, Tamil Nadu (NSHIP, TN)

- **First-of-its-Kind SPV**: Established as India's first dedicated **Special Purpose Vehicle (SPV)** under the Government of India's **Shipbuilding Development Scheme**.
- **Institutional Structure**: 50:50 Joint Venture between **V.O. Chidambaranar (VOC) Port Authority** (Government of India) and **SIPCOT** (State Industries Promotion Corporation of Tamil Nadu, Govt of Tamil Nadu).
- **Strategic Vision**: Develop a mega shipbuilding and heavy industries manufacturing cluster to capture commercial merchant vessel orders and domestic defence vessel repair.

HAL Dhruv New Generation (NG) Helicopter for Civil Aviation

- **Maiden Flight**: Successfully conducted on **30 December 2025** at HAL's Helicopter Division in **Bengaluru**.
- **Technical Specifications**: 5.5-tonne, indigenously developed, twin-engine, all-weather multi-role helicopter derived from the Dhruv Advanced Light Helicopter (ALH) Mk-III civil platform.
- **Propulsion & Certification**: Powered by indigenous **Shakti turboshaft engines** with Cat-A performance and AS4 certification; specifically configured for demanding offshore oil and gas logistics (such as ONGC offshore platforms).

DRAVYA Portal (Ayush Ayurvedic Database)

- **Full Form**: **Digitized Retrieval Application for Versatile Yardstick of AYUSH Substances (DRAVYA)**.
- **Publisher**: Ministry of Ayush.
- **Repository Scope**: Largest comprehensive digital database of Ayurvedic ingredients, classical medicinal plants, and modern products, linking classical Sanskrit ayurvedic texts with peer-reviewed pharmacological and toxicological field studies.
- **Phase I Target**: Catalogues standardized technical dossiers on **100 key medicinal substances**.

Rudrastra Hybrid VTOL Strike UAV

- **Successful Firing**: Tested by the Indian Army on **11 June 2025** at the **Pokhran Firing Range, Rajasthan**.
- **Manufacturer**: Developed indigenously by **Solar Defence and Aerospace Limited (SDAL)**.
- **Platform Architecture**: Hybrid **Vertical Take-Off and Landing (VTOL)** unmanned aerial vehicle, merging the runway-independent vertical launch of a helicopter with the high-speed cruise efficiency of a fixed-wing aircraft.
- **Tactical Parameters**: Total operational range of **170 km** (including target loitering); flight endurance of **~90 minutes**; armed with an **airburst warhead** for counter-insurgency and tactical battlefield interdiction.

SCOT — World's First Commercial Space Surveillance Satellite

- **Pioneering Startup**: Commissioned by Bengaluru-based spacetech enterprise **Digantara**.
- **Payload Name**: **SCOT (Space Camera for Object Tracking)**.
- **Launch Details**: Launched into Low Earth Orbit (LEO) aboard **SpaceX's Transporter-12 mission** on **14 January 2025**; operational imaging initiated in March 2025.
- **Resolution & Capabilities**: World's first commercial orbital surveillance sensor capable of detecting and tracking space debris and orbital objects as small as **5 cm** in LEO, safeguarding satellite constellations against catastrophic orbital collisions.

BSNL Soft Launches Quantum 5G FWA

- **Debut Date & Venue**: Announced on **18 June 2025** in Hyderabad.
- **Innovation**: Bharat Sanchar Nigam Limited (BSNL) soft-launched **Quantum 5G Fixed Wireless Access (FWA)**.
- **Technology Profile**: Indigenous, **SIM-less** fixed-wireless broadband solution delivering gigabit fibre-like speeds wirelessly over 5G radio frequencies to homes and remote enterprises.

India International Science Festival (IISF) 2025 (11th Edition)

- **Dates & Venue**: Held from **6 to 9 December 2025** in **Panchkula, Haryana**.
- **Theme**: *"Vigyan Se Samriddhi: for Atmanirbhar Bharat"* (Science for Prosperity: for a Self-Reliant India).
- **Core Mission**: Bridging traditional knowledge systems with state-of-the-art scientific research and laboratory-to-society technology transfer.

WAVES Animation Bazaar & 8th India Joy 2025 (Hyderabad)

- **Dates & Venue**: Held on **1 November 2025** in **Hyderabad**.
- **Organizers**: Jointly organized by Government of Telangana, TVAGA (Telangana VFX, Animation and Gaming Association), and the Union Ministry of Information & Broadcasting.
- **Strategic Announcement**: Announced the establishment of the **Indian Institute of Creative Technologies (IICT)** in Hyderabad to serve as the premier national institute for AVGC-XR (Animation, Visual Effects, Gaming, Comics, and Extended Reality).

Project Samudrayaan & Deep Ocean Mission (MATSYA-6000)

- **Deep Ocean Dives**: Two Indian aquanauts — **Commander (Retd) Jatinder Pal Singh** and **R. Ramesh** — successfully conducted preparatory deep-ocean training dives into the Atlantic Ocean aboard France's research submersible ***Nautile***.
- **Mission Identity**: **Samudrayaan** is India's first manned deep-sea exploration mission, spearheaded by the Ministry of Earth Sciences (MoES) under the flagship **Deep Ocean Mission**.
- **Manned Vehicle**: **MATSYA-6000**, a deep-submergence spherical vehicle designed to carry **3 humans** to an extreme ocean depth of **6,000 metres (6 km)**.
- **Engineering Specifications**:
- Hull Geometry: Perfect sphere of **2,260 mm internal diameter** with **80 mm thick walls**.
- Metallurgy: Forged from ultra-high-strength **Titanium alloy (Ti6Al4V ELI grade)**.
- Pressure Tolerance: Engineered to withstand crushing hydrostatic pressures up to **600 bar** (600 times atmospheric pressure) at near-freezing ocean temperatures of -3°C.
- **Strategic Objectives**: Underwater scientific exploration, polymetallic nodules / gas hydrates survey, deep-sea mining technology, and benthic ocean robotics.

ICGS Samudra Pratap — India's First Indigenous Pollution Control Vessel

- **Commissioning**: Formally commissioned by Raksha Mantri **Rajnath Singh** on **5 January 2026** in Goa.
- **Shipbuilding & Fleet Standing**: Built by **Goa Shipyard Limited (GSL)**; the first of two dedicated Pollution Control Vessels (PCVs); boasts over **60% indigenous content** and stands as the **largest vessel in the Indian Coast Guard (ICG) fleet to date**.
- **Operational Profile**: Engineered with state-of-the-art ocean containment booms, oil recovery skimmers, chemical dispersant spraying gear, dynamic positioning systems, and firefighting capabilities to neutralize offshore marine ecological disasters.

16th Assembly of the International Renewable Energy Agency (IRENA)

- **Dates & Venue**: Concluded on **12 January 2026** in **Abu Dhabi, United Arab Emirates (UAE)**.
- **Assembly Theme**: *"Powering Humanity: Renewable Energy for Shared Prosperity"*.
- **Youth Leadership**: Convened the IRENA Youth Forum with delegates from over 60 countries to advance digital technology and youth entrepreneurship in the clean energy transition.

Green Mobility Initiatives in Hansalpur (Gujarat)

- **Event Date**: **26 August 2025** in **Hansalpur, Ahmedabad district, Gujarat**.
- **Inaugurated by**: Prime Minister **Narendra Modi**.
- **Technological Breakthroughs**: 1. Inaugurated localized commercial manufacturing of advanced **hybrid battery electrodes** at the **TDS Lithium-Ion Battery Plant**. 2. Flagged off global exports of Suzuki's first Made-in-India global Battery Electric Vehicle (BEV) — the **"e-VITARA"**.

October 2026

India's 2nd Naval VLF Communication Station | Damagundam, Telangana

What happened On **15 October 2026**, Union Defence Minister **Rajnath Singh**, alongside Telangana Chief Minister **A. Revanth Reddy**, laid the foundation stone for the Indian Navy's second **Very Low Frequency (VLF) Communication Transmission Station** at the **Damagundam Reserve Forest** in Vikarabad district, Telangana.

Core specs & prelims facts - **Location:** Damagundam Reserve Forest, Pudur Mandal, **Vikarabad district, Telangana**. - **Outlay & Area:** Estimated cost of approximately **₹3,200 crore**, spanning across **2,900 acres**. - **Operational Mandate:** Operates in the VLF electromagnetic band (3 kHz to 30 kHz). Because VLF radio waves have the unique physical property of penetrating seawater down to dozens of metres, this station provides secure, uninterrupted, encrypted command-and-control communication to deeply submerged nuclear (SSBN/SSN) and conventional (SSK) submarines across the Indian Ocean and Indo-Pacific without requiring them to surface or deploy vulnerable floating wire antennas.

> **UPPCS Confused-Pair Alert**

1. **First vs Second:** Damagundam is India's **SECOND** VLF station.
2. **First VLF Station: INS Kattabomman**, located at Vijayanarayanam in **Tirunelveli district, Tamil Nadu** (commissioned in 1990).
3. Do not confuse VLF (Very Low Frequency for submarines) with ELF (Extremely Low Frequency) or HF (High Frequency).

Exercise Malabar 2026 | Visakhapatnam & Bay of Bengal | 8–18 October 2026

What happened The Indian Navy hosted the **Exercise Malabar 2026**, bringing together all four **Quad** partners—**India, United States, Japan, and Australia**—for high-intensity multilateral maritime drills off **Visakhapatnam** under the aegis of the Eastern Naval Command.

Key assets & participating nations - **Format:** Harbour phase at **Visakhapatnam** (8–12 Oct) followed by Sea phase in the **Bay of Bengal** (13–18 Oct). - **India (Host):** Guided-missile destroyers, multi-mission stealth frigates, submarines, P-8I maritime reconnaissance aircraft, and MARCOS commandos. - **United States:** Arleigh Burke-class guided-missile destroyer *USS Dewey*, integral helicopters, and P-8A Poseidon patrol aircraft. - **Australia:** Anzac-class frigate *HMAS Stuart*, MH-60R helicopter, and P-8 maritime patrol aircraft. - **Japan:** Murasame-class destroyer *JS Ariake*. - **Tactical Focus:** Advanced Anti-Submarine Warfare (ASW), air defence tracking, coordinated fleet manoeuvres, surface live-firings, and Special Forces maritime interdiction.

Global Space Milestones | October 2026

SpaceX Starship Flight 5 — Historic Mid-Air Booster Catch (13 October 2026)

- **What happened:** In a historic aerospace engineering breakthrough at Starbase in Boca Chica, Texas, SpaceX successfully launched **Starship Flight 5** and caught the massive 71-metre-tall **Super Heavy rocket booster** out of mid-air upon its descent using the dual mechanical arms ("chopsticks") of the launch tower (**Mechazilla**).
- **Significance:** Marks the world's first successful mechanical recovery of a super-heavy booster at the launch mount without landing legs, opening the path to rapid, same-day rocket reuse.

NASA Launches Europa Clipper to Jupiter's Moon (14 October 2026)

- **What happened:** NASA successfully launched its flagship **Europa Clipper** spacecraft atop a SpaceX **Falcon Heavy** rocket from Launch Complex 39A at Kennedy Space Center, Florida.
- **Target: Europa**, the icy moon of **Jupiter**, which harbours a global liquid ocean beneath its frozen crust containing more than twice the water of Earth's oceans combined.
- **Mission:** Investigate the icy moon's habitability, ice-shell thickness, ocean chemistry, and geology (scheduled to reach Jupiter orbit in 2030).

Practice Zone — UPPCS Format

> **20 questions** for this sheet only — drill after you revise the months above. Formats match UPPCS 2024–25: multi-statement, Match List, NOT-matched, A/R, chronology. Full cross-topic index → Practice Zone hub.

Q1. With reference to the Prototype Fast Breeder Reactor (PFBR) at Kalpakkam, which of the following statements is/are correct?

1. It achieved first criticality in April 2026.
2. First criticality was achieved using thorium as the primary fuel.

Select the correct answer from the code given below:

A. Only 2 B. Neither 1 nor 2 C. Both 1 and 2 D. Only 1

▼ **Show answer**

Logic: PFBR Kalpakkam — MOX vs thorium trap (science false second sentence). **Ans: D.** Statement 1 is true. Statement 2 is false — criticality used **MOX** fuel; thorium belongs to a later stage of India's three-stage nuclear plan. Trap: equating "Stage-2 bridge to thorium" with "already running on thorium".

Q2. With reference to NavIC / IRNSS developments in March 2026, which of the following statements is/are correct?

1. NavIC is India's regional navigation satellite system.
2. Failure of the atomic clock on IRNSS-1F reduced the number of positioning satellites below the minimum of four.

Select the correct answer from the code given below:

A. Only 2 B. Neither 1 nor 2 C. Both 1 and 2 D. Only 1

▼ **Show answer**

Logic: NavIC vs GPS identity. **Ans: C.** Both are correct facts. Trap: calling NavIC a substitute name for US GPS.

Q3. Which of the following pairs is/are **NOT** correctly matched?

1. Project Kusha — long-range air-defence SAM
2. LR-AShM — hypersonic anti-ship missile displayed around Republic Day 2026
3. Project Kusha — Agni-series ballistic missile with MIRV

Select the correct answer from the code given below:

A. Only 3 B. 1 and 2 C. 2 and 3 D. Only 1

▼ **Show answer**

Logic: Kusha is SAM / air defence, not Agni-class ballistic. **Ans: A.** Pair 3 is wrong. Kusha is air-defence; Agni MIRV is a separate May 2026 defence story. Pairs 1 and 2 are correct.

Q4. Match List-I with List-II and select the correct answer using the code given below the lists.

List-I	List-II
A. INS Aridhaman	1. First privately developed Indian orbital-class rocket (Vikram-1)
B. NASM-SR	2. Third indigenous SSBN
C. Mission Aagaman	3. Indigenous air-launched anti-ship missile (~55 km class)
D. AK-203 'Sher'	4. Fully indigenous rifle production at Korwa, Amethi

Row order is not the answer code.

A. 2 3 1 4 B. 2 1 3 4 C. 3 2 1 4 D. 2 3 4 1

▼ **Show answer**

Logic: Match missile / platform ↔ identity. **Ans: A.** Aridhaman→2; NASM-SR→3; Aagaman/Vikram-1→1; Sher→4. Trap: swapping Aridhaman with a diesel Project-75 boat, or putting Sher in Kanpur/Jhansi.

Q5. With reference to the Defence Acquisition Council decision around 27 March 2026, which of the following statements is/are correct?

1. Acceptance of Necessity was reported at about ₹2.38 lakh crore.
2. The package included more S-400 systems, transport aircraft, Ghatak drones and Dhanush guns in coverage.

Select the correct answer from the code given below:

A. Only 2 B. Neither 1 nor 2 C. Both 1 and 2 D. Only 1

▼ **Show answer**

Logic: DAC money + platform names. **Ans: C.** Big money + named platforms. Trap: swapping the figure with ECLGS 5.0 (~₹2.55 lakh crore credit) from May.

Q6. With reference to LR-AShM displayed around Republic Day 2026, which of the following statements is/are correct?

1. It is a DRDO hypersonic anti-ship missile.
2. Its range in coverage is about 1,500 km.
3. It is an air-defence SAM identical to Project Kusha.

Select the correct answer from the code given below:

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

▼ **Show answer**

Logic: LR-AShM range / nature. **Ans: A.** (3) mixes LR-AShM with Kusha. Trap: writing 55 km (NASM-SR class) as LR-AShM range.

Q7. With reference to naval platforms commissioned / covered around 3 April 2026, which of the following statements is/are correct?

1. INS Aridhaman is the third indigenous SSBN.
2. INS Taragiri is the fourth Project 17A frigate in coverage.
3. Both events were linked with Visakhapatnam in coverage.

Select the correct answer from the code given below:

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

▼ **Show answer**

Logic: INS Taragiri + Aridhaman same day cluster. **Ans: D.** SSBN vs frigate — do not merge into one class. Trap: calling Aridhaman a diesel Project-75 boat.

Q8. With reference to the advanced Agni MIRV test (8 May 2026), which of the following statements is/are correct?

1. The flight trial was from Dr APJ Abdul Kalam Island, Odisha.
2. MIRV means one missile can deliver multiple warheads against multiple targets.
3. The same test is identical to the Project Kusha SAM maiden flight.

Select the correct answer from the code given below:

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

▼ **Show answer**

Logic: Agni MIRV test place / nature. **Ans: A.** (3) false — Kusha is July air-defence SAM. Same island, different missile family.

Q9. Match List-I with List-II and select the correct answer using the code given below the lists.

List-I	List-II
A. INS Dunagiri	1. Coast Guard ship in June coverage
B. ICGS Akshay	2. Navy platform in June coverage
C. INS Mahendragiri	3. Sixth Project 17A frigate (July coverage)
D. INS Nipun	4. Second Nistar-class diving support vessel (August coverage)

Row order is not the answer code.

A. 2 1 3 4 B. 1 2 3 4 C. 2 1 4 3 D. 2 3 1 4

▼ **Show answer**

Logic: Navy June platforms Match. **Ans: A.** Dunagiri→Navy June; Akshay→ICG; Mahendragiri→6th 17A; Nipun→Nistar-class. Trap: putting all under one service.

Q10. With reference to National Space Day 2026 (23 August), which of the following statements is/are correct?

1. It was the 3rd edition of National Space Day.
2. Coverage spoke of about 8 launch-vehicle missions and 12 satellite missions planned for the financial year.
3. The day is observed on 15 August every year.

Select the correct answer from the code given below:

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

▼ **Show answer**

Logic: National Space Day theme / mission counts. **Ans: A.** (3) false — National Space Day is **23 August** (Chandrayaan-3 landing anniversary logic). Trap: merging with I-Day.

Q11. Consider the following events and arrange them in correct chronological order:

1. Project Kusha maiden test
2. Advanced Agni MIRV test
3. PFBR Kalpakkam first criticality
4. Vikram-1 / Mission Aagaman launch window

Select the correct answer from the code given below:

A. 3, 2, 4, 1 B. 2, 3, 4, 1 C. 3, 2, 1, 4 D. 3, 4, 2, 1

▼ **Show answer**

Logic: Chronology — defence / science firsts. **Ans: A.** PFBR 6 Apr → Agni MIRV 8 May → Vikram-1 18 Jul → Kusha 23 Jul. Trap: putting Kusha before Vikram because both are July.

Q12. With reference to Exercise Pragati 2026, which of the following statements is/are correct?

1. It was held at Umroi, Meghalaya.
2. Coverage spoke of about 13 participating nations.
3. It was a purely bilateral India–France naval drill in the Arabian Sea.

Select the correct answer from the code given below:

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

▼ **Show answer**

Logic: Exercise Pragati 2026. **Ans: A.** (3) invents a different exercise. Trap: recycling Varuna/Malabar venues.

Q13. With reference to May 2026 missile tests, which of the following statements is/are correct?

1. An advanced Agni MIRV test was conducted on 8 May from Kalam Island.
2. An Agni-1 test under Strategic Forces Command was also covered around 22 May.
3. Both tests were Project Kusha SAM flights.

Select the correct answer from the code given below:

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

▼ **Show answer**

Logic: Agni-1 under SFC May. **Ans: A.** (3) Kusha is July.

Q14. With reference to NASM-SR, which of the following statements is/are correct?

1. It is an indigenous air-launched anti-ship missile.
2. Range in coverage is about 55 km.
3. Range in coverage is about 1,500 km.

Select the correct answer from the code given below:

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

▼ **Show answer**

Logic: NASM-SR range and nature. **Ans: A.** (3) is LR-AShM range.

Q15. Which of the following statements is/are correct for June 2026 naval coverage?

1. INS Dunagiri appeared among Navy platforms inducted/covered.
2. ICGS Akshay is a Coast Guard ship in the same window.
3. ICGS Akshay is a Project 17A frigate.

Select the correct answer from the code given below:

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

▼ **Show answer**

Logic: INS Dunagiri / Sanshodhak / Agray June navy. **Ans: A.** (3) service mix-up.

Q16. With reference to Project Kusha coverage, which of the following statements is/are correct?

1. It fits the wider Mission Sudarshan Chakra air-defence shield idea.
2. Induction window spoken of around 2028–2030.

3. It is a ballistic MIRV Agni variant.

Select the correct answer from the code given below:

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

▼ **Show answer**

Logic: Mission Sudarshan Chakra / Kusha induction window. **Ans: A.** (3) wrong family.

Q17. With reference to ACV H-561, which of the following statements is/are correct?

1. It is an Air Cushion Vehicle for the Coast Guard.
2. Coverage called it the first indigenous ACV of its kind for ICG.

Select the correct answer from the code given below:

A. Only 2 B. Neither 1 nor 2 C. Both 1 and 2 D. Only 1

▼ **Show answer**

Logic: ACV H-561 Coast Guard first indigenous. **Ans: C.** Both true.

Q18. Which of the following statements is/are correct?

1. January coverage included Pralay salvo / test stories.
2. K-4 sea-deterrent related tests also appeared in January defence CA.
3. K-4 is a Ramsar wetland in UP.

Select the correct answer from the code given below:

A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3

▼ **Show answer**

Logic: Pralay / K-4 January soft. **Ans: A.** (3) absurd distractor.

Q19. The PFBR at Kalpakkam is locked in coverage as which capacity class?

A. 100 MWe B. 500 MWe C. 700 MWe D. 1000 MWe

▼ **Show answer**

Logic: PFBR power rating. **Ans: B.** 500 MWe fast breeder.

Q20. With reference to INS Nipun (August 2026), which of the following statements is/are correct?

1. It is the second Nistar-class diving support vessel in coverage.
2. It is the sixth Project 17A frigate.

Select the correct answer from the code given below:

A. Only 2 B. Neither 1 nor 2 C. Both 1 and 2 D. Only 1

▼ **Show answer**

Logic: INS Nipun Nistar-class. **Ans: D.** Mahendragiri is sixth 17A.

Practice Zone — UKPCS & Multi-State Science/Defence Drill

> **5 questions** targeting joint military exercises, ISRO missions, and high-altitude warfare nodes relevant for both state PSCs.

Q21. Exercise 'Surya Kiran' is a bilateral joint military exercise conducted between the armed forces of India and which neighbouring country?

A. Bhutan
B. Nepal
C. Bangladesh
D. Sri Lanka

▼ **Show answer**

Logic: Surya Kiran is an annual joint military exercise between the Indian Army and the Nepalese Army, frequently hosted at Pithoragarh in Uttarakhand and Saljhandi in Nepal. **Ans: B.** Nepal.

Q22. With reference to the historic lunar landing of Chandrayaan-3 on 23 August 2023, which of the following statements is/are correct?

1. The touchdown point of the Vikram lander near the lunar South Pole was officially named ‘Shiv Shakti Point’.
2. India became the fourth country in the world to achieve a soft lunar landing, and the first near the lunar South Pole.
3. August 23 is now commemorated annually across India as ‘National Space Day’.

Select the correct answer from the code given below:

A. 1 and 2 only
B. 2 and 3 only
C. 1 and 3 only
D. 1, 2 and 3

▼ **Show answer**

Logic: All three statements are landmark facts of Chandrayaan-3 (launched aboard LVM3-M4). The Chandrayaan-2 impact site was named 'Tiranga Point'. **Ans: D.** 1, 2 and 3.

Q23. Which town in the Chamoli district of Uttarakhand hosted the Indo-US joint military exercise ‘Yudh Abhyas’ for high-altitude cold-climate warfare training?

A. Joshimath
B. Auli
C. Badrinath
D. Gopeshwar

▼ **Show answer**

Logic: The Indo-US joint military exercise 'Yudh Abhyas' was conducted in the snow-clad slopes of Auli (Chamoli district, Uttarakhand), emphasizing high-altitude, cold-weather warfare techniques. **Ans: B.** Auli.

Q24. Which primary scientific payload aboard India’s solar mission Aditya-L1 was designed and developed by the Indian Institute of Astrophysics (IIA), Bengaluru?

A. SUIT
B. VELC (Visible Emission Line Coronagraph)
C. PAPA
D. ASPEX

▼ **Show answer**

Logic: The Visible Emission Line Coronagraph (VELC) is the primary internal coronagraph payload of Aditya-L1, developed by IIA Bengaluru to image the solar corona continuously down to 1.05 solar radii. **Ans: B.** VELC.

Q25. Joint military exercise ‘Dustlik’ is conducted between the armies of India and which Central Asian nation?

A. Kazakhstan
B. Uzbekistan
C. Tajikistan
D. Kyrgyzstan

▼ **Show answer**

Logic: Exercise Dustlik is a bilateral army exercise between India and Uzbekistan, with editions hosted at Pithoragarh in Uttarakhand and Termez in Uzbekistan. **Ans: B.** Uzbekistan.